

Configuring Circuit Breaker on Tyk OAS API

Configuring Circuit Breakers in Tyk OAS Definitions

Leverages OpenAPI Operation IDs

- Tyk attaches the circuit breaker to specific operationIds declared in your OpenAPI document.
- These are linked to method + path, e.g., GET /status/200 → status/200get.

Supports Wildcards

- OpenAPI paths like /status/{code} are treated as wildcards → matched as (.*)
- Easier to configure human-readable paths while applying policies broadly.

Defined in x-tyk-api-gateway.middleware.operations

```
"middleware": {
  "operations": {
    "status/200get": {
      "circuitBreaker": {
        "enabled": true,
        "threshold": 0.5,
        "sampleSize": 10,
        "coolDownPeriod": 60,
        "halfOpenStateEnabled": true
      }
    }
  }
}
```

Circuit Breaker Parameters

Parameter	Purpose
<code>enabled</code>	Turns the middleware on or off for this operation
<code>threshold</code>	The error rate limit (0.0–1.0) that trips the breaker
<code>sampleSize</code>	Minimum # of requests in the 10s rolling window before breaker can trip
<code>cooldownPeriod</code>	How long (in seconds) the breaker remains open once tripped
<code>halfOpenStateEnabled</code>	Enables probing mode to restore service before cooldown ends

Example – Full Tyk OAS API Definition

```
{
  "paths": {
    "/status/200": {
      "get": {
        "operationId": "status/200get"
      }
    }
  },
  "x-tyk-api-gateway": {
    "middleware": {
      "operations": {
        "status/200get": {
          "circuitBreaker": {
            "enabled": true,
            "threshold": 0.5,
            "sampleSize": 10,
            "coolDownPeriod": 60,
            "halfOpenStateEnabled": true
          }
        }
      }
    }
  }
}
```

Demo:
**Configuring Circuit
Breaker on Tyk
Dashboard**